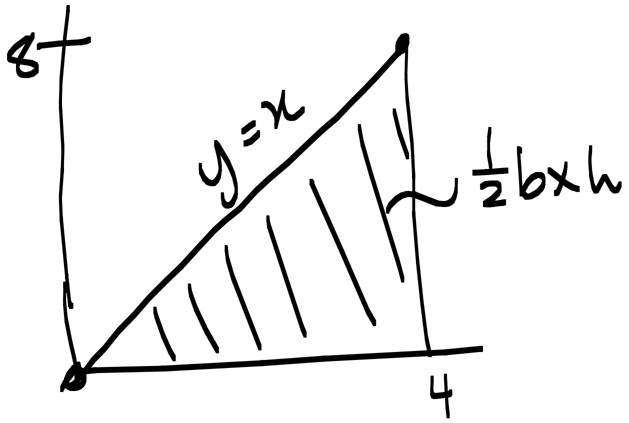
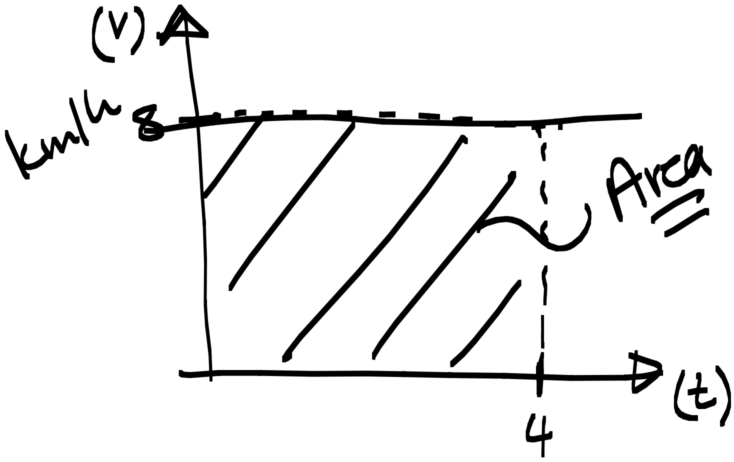
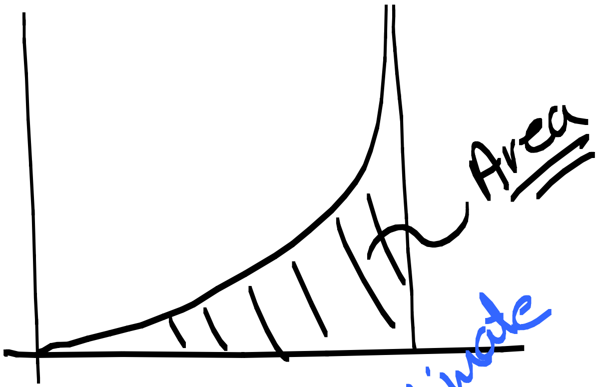
Unit 5.2 The area problem p366

Note Title 2014/04/20





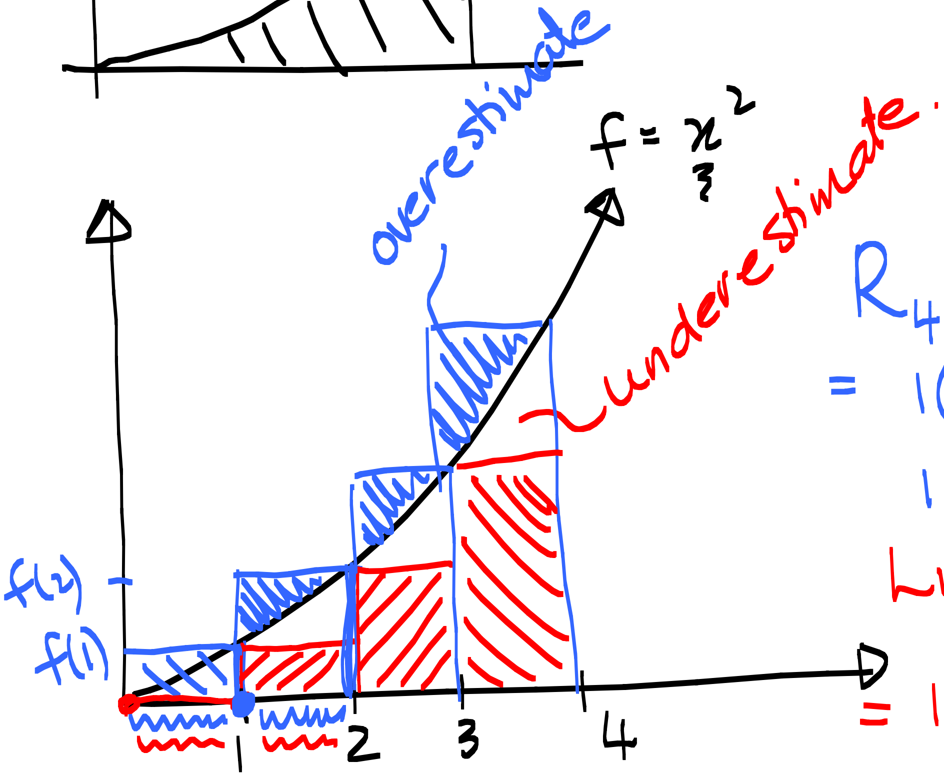


Travel for 4 hours at 8 km/h.

$D = 8 \times 4$

$\quad km/h \times h$

$= 32$ km



R_4

$= 1(f(1)) + 1(f(2)) + 1(f(3)) + 1(f(4))$

L_4

$= 1(f(0)) + 1\,f(1) + 1\,f(2) + 1\,f(3)$

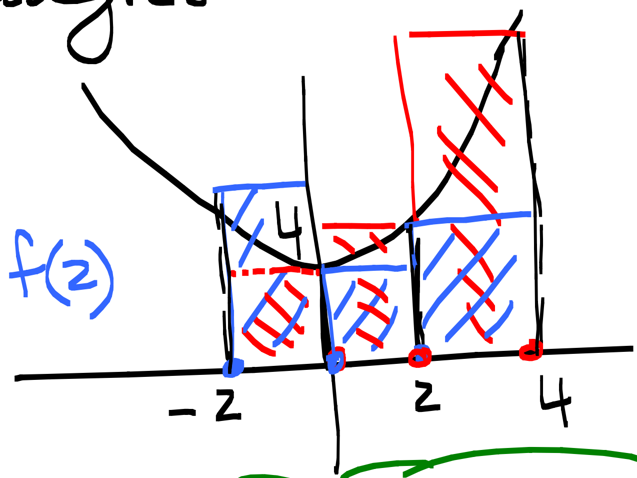
$R_4 = 1(1) + 1(4) + 1(9) + 1(16) = 30$

$L_4 = 14$

$\text{Area} \approx \text{Avg} = \dfrac{30+14}{2}$

$= 22$

1. Find the area under the curve of $f(x) = x^2 + 4$ on $[-2, 4]$ using 3 rectangles



length of intervals = 2

Left hand sum (LHS)

$$\text{Area} \approx 2f(-2) + 2f(0) + 2f(2)$$
$$= 2(8) + 2(4) + 2(8)$$
$$= 40$$

Right hand sum (RHS)

$$\text{Area} \approx 2f(0) + 2f(2) + 2f(4) = 64$$

$$40 \leq \text{area} \leq 64$$

x	-2	0	2	4
$f(x)$	8	4	8	20

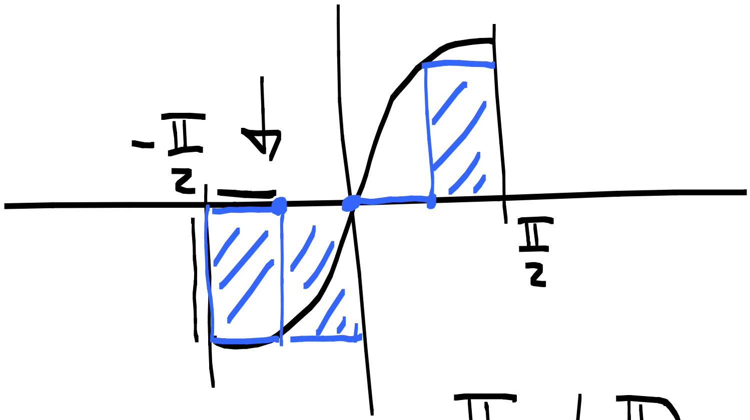
Overestimate – use the largest function value in each interval.

Underestimate – use the smallest function value in each interval.

$$\text{Over} \approx 2(8) + 2(8) + 2(20) = 72$$

$$\text{Under} \approx 2(4) + 2(4) + 2(8) = 32$$

2. Use 4 rectangles to find the Right hand sum of $f(x) = \sin x$ on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$



Size of intervals: $\dfrac{\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)}{4} = \dfrac{\pi}{4}$

x	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
$f(x)$	-1	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$	1

$$LHS = \frac{\pi}{4}(-1) + \frac{\pi}{4}\left(-\frac{1}{\sqrt{2}}\right) + \frac{\pi}{4}(0) + \frac{\pi}{4}\left(\frac{1}{\sqrt{2}}\right)$$

$$= -\frac{\pi}{4}$$

If a function f is increasing on an interval $[a, b]$ will the LHS be an over or under estimation of the area under the curve?